POLITECNICO
MILANO 1863

Dipartimento di Elettronica e Informazione e Bioingegneria

Advanced Computer Architectures

September 03, 2025

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Name
Last Name
ID Number (codice persona)

Problem 1 (10%)	
Problem 2 (15%)	
Problem 3 (25%)	
Question 1 (15%)	
Question 2 (20%)	
Question 3 (15%)	
Total (100%)	

NOTE: students have 90' to complete the exam.
To pass the midterm, a minimum score of 20% is requested by each "section" (section 1: problems, section 2: questions).
The overall score to access the oral has to be greater or equal to 50%
English is the only language allowed for the answers

Problem 1 (10%)

Describe (the answer has to be effectively supported) a 1-BHT and a 2-BHT able to execute the following assembly code

Assembly:

```
I0: li $R1, 0
I1: li $R2, 8000
FOR:LD $F2, A($R1)
I2: LD $F4, B($R1)
I3: LD $F6, C($R1)
I4: FADD $F4, $F2, $F4
I5: FADD $F6, $F4, $F6
I6: SD $F6, A($R1)
I7: ADDUI $R1, $R1, 8
I8: BNE $R1, $R2, FOR
```

P1.A The obtained result, in terms of miss predictions, is in line with theoretical characteristics of the two predictors? Please effectively support your answer.

P1.B List all the static branch predictors you know and discuss their performance in terms of mispredictions, if possible.

P1.A Answer

P1.B

Problem 2 (15%)

Consider the following access pattern on a three-processor system with a direct-mapped, write-back cache with one cache block and a three cache block memory.

P2.A Is this a possible pattern with MESI, with write-back caches, write-allocate, and write-invalidate of other caches? Write also a justification.

Time	After Operation	Px cache block state	Py cache block state	Pz cache block state	Memory at Block 0 up to date?	Memory at Block 1 up to date?	Memory at Block 2 up to date?
1	Px read block 2	Exclusive (0)	Shared (2)	Shared (2)	yes	yes	yes
2	Pz write block 2	Exclusive (0)	Exclusive (2)	Modified (2)	yes	yes	no
3	Px read block 2	Shared (2)	Shared (2)	Modified (2)	yes	yes	no
4	Pz read block 1	Shared (2)	Shared (2)	Exclusive (1)	yes	yes	yes
5	Py write block 2	Invalid (2)	Modified (2)	Exclusive (1)	yes	yes	no
6	Px read block 2	Shared (2)	Shared (2)	Exclusive (1)	yes	yes	yes
7	Py write block 1	Exclusive (2)	Modified (1)	Invalid (1)	yes	no	yes
8	Pz read block 0	Exclusive (2)	Modified (1)	Exclusive (0)	yes	no	yes
9	Px write block 0	Modified (0)	Modified (1)	Invalid (0)	no	no	yes
10	Py read block 2	Modified (0)	Exclusive (2)	Invalid (0)	no	yes	yes
11	Py read block 2	Modified (0)	Exclusive (2)	Invalid (0)	no	yes	yes
12	Pz write block 1	Modified (0)	Exclusive (2)	Modified (1)	no	no	yes
13	Px read block 1	Shared (1)	Exclusive (2)	Shared (1)	yes	yes	yes
14	Pz read block 0	Exclusive (1)	Exclusive (2)	Exclusive (0)	yes	yes	yes
15	Pz read block 1	Shared (1)	Exclusive (2)	Shared (1)	yes	yes	yes
16	Py read block 1	Shared (1)	Shared (1)	Shared (1)	yes	yes	yes
17	Px write block 1	Modified (1)	Invalid (1)	Invalid (1)	yes	no	yes
18	Px read block 1	Modified (1)	Invalid (1)	Invalid (1)	yes	no	yes
19	Pz write block 1	Invalid (1)	Invalid (1)	Modified (1)	yes	no	yes

P2.B Compute the miss rate of the previous execution pattern. Consider a miss all the times the required block is not already in cache.
Consider a hit time of 10 ns and a miss penalty of 50ns. Compute the average access time for Px. What would you change in the memory subsystem to increase the average access time?

Answer P2.A

Answer P2.B

Problem 3 (25%)
Consider the following assembly executed to be executed on a CPU with dynamic scheduling based on TOMASULO with a ROB as shown below.

```
    li t0, 0
L1: li   t6, 2
    mul  t3, t0, t6
    li t1, 0
L2:  add t4, t0, t1
     addi t1, t1, 1
     li   t2, 100
     blt  t1, t2, L2
    mul t5, t0, t0
    addi t0, t0, 1
    li   t2, 1000
    blt  t0, t2, L1
```

	Instruction	ISSUE	START EXE	EXE COMPLETE	WB	Commit	ROB idx	RSi	Unit	Hazards
1	`li t0, 0`	1	2	4	5	6				
2	`L1: li   t6, 2`	2	3	5	6	7				
3	`mul  t3, t0, t6`	3	7	12	13	14				
4	`li t1, 0`	4	5	7	8	15				
5	`L2:add t4, t0, t1`	5	9	11	12	16				
6	`addi t1, t1, 1`	6	9	9	10	17				
7	`li   t2, 100`	7	8	10	11	18				
8	`blt  t1, t2, L2`	8	12	12	13	19				
9	`mul t5, t0, t0`	9	10	15	16	FLUSHED				
10	`addi t0, t0, 1`	10	11	11	12	FLUSHED				
11	`li   t2, 1000`	11	12	14	15	FLUSHED				
12	`blt  t0, t2, L1`	15	16	16	17	FLUSHED				

Assumptions: I8 branch is predicted not taken, thus it executes I9 sequentially. The misprediction is recognized at commit time, after which all the wrongly speculated instructions are flushed. Integer operations have a separate CDB than Mem/FP operations.

P3. A Consider true data dependencies only. List all of the them. Are they honored in the table above? Please add a "+" for each dependency that turns into hazard in the "Hazard" column.
Answer P3.A

P3. B Consider name dependencies only. What they are? Are they honored in the table above? How? Please add a "*" for each dependency that turns into hazard in the "Hazard" column.
Answer P3.B

P3. C Consider control dependencies and structural hazards. dependencies only. What they are? Are they honored in the table above? How? Please add a "•" for each dependency that turns into hazard in the "Hazard" column. Fill also the "ROB idx", "RSi", and "Unit" columns appropriately.

Answer P3.B

P3. C Is there a Tomasulo configuration that could respect the shown execution? If yes, please describe ALL the architectural requirements to be compatible with the table and the assumptions. If no, please motivate your answer.

Answer P3.C

P3. D Calculate the CPI of this architecture (ARCH1) for the given execution table (regardless it works or not).
Answer P3.D

P3. E Consider a frequency of 700 MHz for ARCH1 and 500MHz for ARCH2. Which CPI should have ARCH2 to be faster than ARCH1?
Answer P3.D